Quantitative Finance, Vol. 12, No. 8, August 2012, 1299–1314

Financial factors and firm growth: evidence from financial data on Taiwanese firms

KHURSHID M. KIANI*†, ELLEN HUIRU CHEN‡
and ZAGROS MADJD-SADJADI§

†Nottingham University Business School, 199 Taikang East Road, Ningbo 315100, China
‡Department of Regulatory Finance, Southern California Edison,
2244 Walnut Grove Ave, Rosemead, CA 91770, USA
§Department of Economics, Winston-Salem State University, Winston-Salem, NC 27110, USA

(*Received 3 June 2007; in final form 17 January 2011*)

We investigate the possible predictability of firm growth in Taiwan using cross-sectional data of financial factors for the years 1997 and 2003 via principal component analysis. Our results reveal that the 18 financial variables (sales growth rate, total assets, total sales, return on assets, return on equity, gross margin, operating cost minus depreciation divided by sales plus other trading income, acid test ratio, debt–equity ratio, time interest earned, average receivables per average daily sales, inventory, average payables per average daily sales, working capital, working capital as a fraction of total assets, long-term liabilities as a fraction of total assets, and sales as a fraction of net worth of the firm) that we employ bunch together into five different financial ratios for the years 1997 and 2003 that are stable between these years. These financial factors are short-term liquidity, return on investment, long-term liquidity, firm size and capital turnover. Regressing these ratio groups (extracted principal components) on firm growth, we find return on investment in the year 1997 was positively and significantly related to firm growth, while long-term solvency was negatively related to firm growth. In addition, smaller firms tended to grow faster. By 2003, larger firms grew faster than smaller ones and short-term liquidity was positively and significantly related to firm growth, while return on investment was no longer a significant determining factor. Our findings suggest that firms that finance internally or do not rely too heavily on indebtedness may end up growing slower during boom periods but they are the ones that survive and outperform after the bust.

Keywords: Applied econometrics; Control and optimization; Corporate finance; Empirical finance

1. Introduction

East Asian economies have attracted the attention of international investors since the 1980s when firms in these economies were viewed as comprehensive and adept at exploiting new investment opportunities. In addition, these economies were targeted by international researchers who were interested in exploring the causes of the so-called ‘Asian Miracle’. However, after the 1997 Asian Financial Crises, international researchers switched to investigate its causes.

Pomerleano (1998) pointed out that a common reason for the unsustainability of high growth in most East Asian economies was excessive investment through borrowing without considering poor returns on investments. Unsustainability of rapid growth in Asian firms could be attributed to arbitrage profits due to changes in market value. A number of theoretical, as well as empirical, models demonstrated that a firm’s value as well as its investment decisions are strongly affected by its capital structure provided the underlying assumptions of the Modigliani–Miller Irrelevance Proposition¶ are violated. For example, capital market imperfection characterized

*Corresponding author. Email: mkkiani@yahoo.com

¶Modigliani and Miller (1958, 1963), Miller and Modigliani (1961), and Miller (1977).

Quantitative Finance
ISSN 1469–7688 print/ISSN 1469–7696 online © 2012 Taylor & Francis
http://www.tandfonline.com
http://dx.doi.org/10.1080/14697688.2011.556143

by asymmetrical information between managers and investors (Arrow 1979) hampers investors' ability to observe all actions taken by managers.

Information asymmetry can generate moral hazard and adverse selection problems both in risk and debt markets (Akerlof 1970). Myers and Majluf (1984) suggested charging a premium for purchasing the shares of stocks of good firms to offset losses because of adverse selection and moral hazard problems. Likewise, credit institutions would be inclined to charge an extra price on new debts or even ration applications for issuing new debts. Yet a key consideration for examining Taiwanese firms is that they tend to be controlled by the owners themselves or by managers who are family friends loyal to the owners and whose interests are strongly interwoven with the interests of the owners. The result is that such information asymmetry issues are reduced. This cultural factor, coupled with the fact that Taiwan better weathered the storm of the East Asian Financial Crisis when compared with other economies in the region, makes the island economy a unique testing ground for financial analysis. After all, the desire not to dilute ownership because of familial desires to maintain control may be at the heart of why Taiwan did not suffer as much as other East Asian countries when the tide turned.

Taiwan may offer lessons for other countries since it has a corporate culture similar to that of the United States, but its lack of well-developed financial markets when compared with other Asian countries has led it to rely on internal financing. Nevertheless, external financing through financial markets becomes more difficult when credit crunches like that in 1997–8 in East Asia and that in 2007–9 in North America and Europe occur. However, to undertake a study of the island economy and generate critical insights for the most recent crisis in the West, we need to focus on the type of variables that would be better suited for investigating firm growth in Taiwan. Unfortunately, most theoretical and empirical studies that have been undertaken so far have focused on widely held† corporations in more developed economies and it is unclear whether the variable selection in such studies are appropriate for investigating an economy where equities tend to be more closely held, such as Taiwan. This paper, therefore, is an exploratory data analysis that seeks to determine, using data reduction via principal component analysis, the components that can be used in a regression analysis to analyse the reasons behind firm growth in Taiwan during the crisis and immediately thereafter.

It is common practice for practitioners and financial managers to analyse a firm's performance using information on financial variables and ratios that are obtained from its balance sheet, income statement, and statement of cash flows. Such traditional financial ratio comparisons are widely used by practitioners in business to check the financial health of business enterprises; however, interpretations of these numbers are subject to individual judgment and arbitrary rules of thumb. These variables are highly correlated with each other, posing the threat of unstable parameter estimates. Various researchers have tried to bridge the gap by linking traditional financial ratio analysis with more rigorous statistical techniques. Most of these studies are focused on bankruptcy prediction and corporate bond ratings. For instance, Beaver (1966) and Pinches and Mingo (1973) worked on bankruptcy prediction, while Altman (1968) extended this analysis using multivariate discriminant analysis. Grice and Ingrim (2001) showed that the predictive powers of the Altman model were flawed, particularly when these models were applied to service firms rather than manufacturing firms.

To obviate the problems associated with collinearity among the variables that are obtained from firm financial statements, Cybinski (2001) used principal component analysis (PCA) to study economic time series to determine whether macroeconomic shocks were responsible for systematic failure of firms, Chen *et al.* (2007b) used PCA to categorize financial ratios into factors that influenced the returns on investment for Chinese A-shares on the Shenzhen and Shanghai stock exchanges, and Shatzberg and Weeks (2004) used PCA to choose between debt or equity issuance. Indeed, PCA is a data-reduction technique that helps to capture all the implied variations in a general pattern of variations while eliminating collinearity between the original variables. These general patterns of variations obtained from such an analysis are eventually employed in regression analysis to forecast firm growth. Considering the foregoing discussion and the fact that no one, to our knowledge, has tried answering this question with respect to a small open economy, we feel that the present study contributes to the literature. We analyse 18 financial variables‡ that were obtained from financial statements of Taiwanese firms using principal component analysis (PCA) as done by others, including Meador *et al.* (1996), Spathis *et al.* (2002), Burkowitz *et al.* (2003), Durnev and Kim (2005), Chan and Mozurkewich (2006), Yusdar *et al.* (2007), and others for solving problems in their areas of interest.

The paper is organized as follows. In section 2, we elaborate the data used, the construction of variables, a discussion of principal component analysis, and the empirical model employed. Section 3 provides the empirical results and a discussion thereof, and section 4 gives the conclusions.

†See Chen *et al.* (2007a) for a discussion of widely held and closely held firms with special emphasis on Taiwan.

‡Sales growth rates, total assets, net sales, return on assets, return on equity, gross margin, operating cost minus depreciation divided by sales plus other trading income (OI %), acid test, debt equity ratio, time interest earned, average receivables per average daily sales, inventory, average payable per average daily sales, working capital, working capital as a fraction of total assets, long-term liabilities as a fraction of total assets, and sales as a fraction of net worth of the firm.

2. Empirical analysis

2.1. Data and definition of variables

The present study employs cross-sectional data for the years 1997 and 2003 for selected Taiwanese firms in order to determine which financial variables (if any) are candidate variables for predicting firm performance and growth. The impact of various patterns of financial variables is also analysed to identify variations in financial data during 1997 and 2003.

Financial data for the year 1997 was selected to study the impact of the '1997 Asian Financial Crisis' on Taiwanese firms since this was the year when the crisis first erupted; therefore, this year can be treated as the base year for the analysis. Likewise, the year 2003 is selected because this was the first year when the Taiwanese economy appeared to stabilize; therefore, this is the year when the analysis can be repeated since, by this year, a sufficient time had elapsed to observe significant changes in the data with respect to the base year.

Cross-sectional data were obtained for 120 good-performing firms from the list published by the Common Wealth Magazine for the years 1997 and 2003, and 25 poorly performing companies from the list published by Taiwan Stock Outlook during the same period. All firms on these two lists were initially included and these firms were drawn from a broad cross-section of industries in the Taiwanese economy. We screened the data to exclude companies with missing observations during the years 1997 and 2003. This resulted in a reduction of the data to 93 firms for the year 1997 and to 114 firms for the year 2003. A response variable and 17 explanatory variables† were selected and compiled from the multitude of available financial ratios for 1997 and 2003. These explanatory variables‡ were used to construct factors to be employed using PCA. Table 1 shows a total of 18 financial ratios for inclusion in the analysis. These financial ratios represent the overall financial dimensions of the Taiwanese firms in a variety of ways, although there is no consistent statistical or logical superiority for particular independent variables among the alternatives, and there is no pre-determined structure or underlying statistical model of the observed variables.

Table 1. Structure of the variables.

Variable	Description
SGR	Sales growth rate (dependent variable)
TA	Total assets
NS	Sales
ROA	Return on assets (ROA = EBIT/total assets)
ROE	Return on equity (ROE = net income/total equity)
GM	Gross margin (GM = gross profit/sales)
OI	OI % (operating costs minus depreciation/total sales plus other trading income)
CR	Current ratio (CR = current assets/current liabilities)
ATR	Acid test (cash plus marketable securities plus receivables/current liabilities or current assets minus stocks/current liabilities)
DER	D/E ratio (L-T debt plus values of leases/total equity)
TIE	Time interest earned (TIE = EBIT plus depreciation/interest earned)
DAR	Days—A/R (DAR = average receivables/average daily sales)
DI	Days—inventory (DI = cost of goods sold/average inventory)
DAP	Days—A/P (DAP = average payable/average daily sales)
WC	Working capital (WC = cash, cash deposits, stocks and debtors, itemized in the funds flow statement. Also known as circulating capital)
WCA	Working capital/total assets
LTL	Long-term liabilities/total assets
SLNW	Sales/net worth

The selection of a large number of financial variables from the financial statements of the firms may be questionable since a few variables can also be used to predict firm growth in Taiwan. Therefore, researchers usually set up criteria for choosing variables according to data availability, existing empirical studies in the area, and financial characteristics that are related to the research design. A number of researchers, including Audretsch and Elston (2002), Carpenter and Petersen (2002), Bond *et al.* (2003), Fagiolo and Luzzi (2006), Oliveira and Fortunato (2006), and Musso and Schiavo (2008), have focused on testing the impact of individual financial factors such as cash-flow and investment decisions on firm growth. By selecting a single or a few ratios from a financial data set, researchers have been trying to identify the underlying theoretical assumptions of the statistical model that would describe the salient characteristics of a firm's activities where each financial ratio conveys unique information about those activities.

In many cases, one might be able to seek guidance from *a priori* theoretical information about the variables of interest to be included in the analysis. However, there can still be a problem of having too many variables to include. The decision of what variables to include becomes purely judgmental, particularly where the relevant theory does not provide appropriate guidance. In the present case, the empirical validity of the financial ratios in the case of Taiwan has not been fully explored§ for the reasons

†The explanatory variables of this study were public information that could be collected and computed from published annual financial statements produced for the Taiwan Stock Exchange Committee (TSEC).

‡Sales growth rates, total assets, net sales, return on assets (ROA), return on equity (ROE), gross margin, OI, acid test, debt equity ratio, time interest earned, average receivables per average daily sales, inventory, average payable per average daily sales, working capital, working capital as a fraction of total assets, long-term liabilities as a fraction of total assets, and sales as a fraction of the net worth of the firm.

§Chen and Shimerda (1981) used principal component analysis in conjunction with more than 100 variables and dozens of financial ratios that were employed in 26 prior studies for bankruptcy prediction and showed that the theory does not provide any basis for selection of these ratios. Likewise, LeClare (2005) noted that the selection of variables is an *ad hoc* process and the unavailability of economic theory as a means of selecting such variables means that although studies cannot be generalized, their specificity leads to a high predictive power for the task at hand.

stated earlier. Therefore, existing theory may not inherently reflect all the relationships among the individual ratios and the presumed statistical model may not be established. Without full knowledge of the empirical relationships existing among individual financial ratios, attempts to draw a meaningful and thoughtful selection of variables are not necessarily achieved. As a result, the ratios chosen are often neither exhaustive nor exclusive in their ability to describe firm behavior that would best represent and keep the most information out of the original financial data set.

Although the selection of the initial variables is still guided by both theory and data availability, we employ principal component analysis (PCA) to isolate dominating dimensions within the data and additionally to overcome serious limitations surrounding the OLS and 2SLS regression techniques when dealing with a large number of possible collinear explanatory variables. Indeed, PCA generates a new set of uncorrelated variables known as principal components, henceforth called components. On the other hand, these components are difficult to interpret, especially when it comes to the interpretation of coefficients. As such, this may be best described as an exploratory study that allows us to see exactly what types of variables have the largest impact on the growth rate of Taiwanese firms as opposed to the magnitude of those impacts. In some cases, variables were chosen because they existed in the sole unified dataset available for these companies, while other variables were not used since those were not accessible. The dataset involved did not include all financial ratios and all of it had to be culled by hand. This time-consuming and expensive process means that the full range of variables that might be found in corporate filings were not available. On the other hand, each of these selected variables either plays an important role in determining firm growth according to one or more studies in the literature or it is closely related to a variable that has been identified by others. The next section discusses the rationale behind the selection of each of these variables.

Total assets, rather than net assets, were used as a denominator for return on assets for the same reason that they were used by Glancey (1998), since net assets are far more variable for smaller firms. Total assets are used in models of firm growth by Chittenden *et al.* (1996) and Diambeidou *et al.* (2008). The variable 'Net Sales' is used to distinguish the level from the growth rates because net sales are not proportional to other indicators of firm size (Smyth *et al.* 1975). Returns on assets as well as return on equity are cited theoretically as being important for firm growth by Demirguc-Kunt and Maksimovic (1998) and Gulati and Zantout (1997). Gross margin is suggested by Wiklund (2006) to be a good estimate of profitability and is related to firm growth. OI% is another profitability measure used by Padachi (2006). The acid test ratio (Hutchinson and Xavier 2006), debt equity ratio (Heshmati 2001), and sales as a fraction of net worth are used to determine credit constraints. The times interest earned ratio and long-term liabilities as a fraction of total assets are included for the same reason. Average receivable per average daily sales ratio, which is also known as the collection ratio, inventory, working capital, and working capital as a fraction of total assets, all fall under the heading of working capital, which is suggested by Carpenter and Petersen (2002) to be an important constraint on growth for small firms.

Smyth *et al.* (1975) have previously shown that various measures of size are not substitutable for one another; the inclusion of several measurements of factors is encouraged when collinearity can be corrected through reliable techniques such as PCA. This allows capturing all of the variance associated with the various measurements and none of the covariance.

2.2. *Preliminary data analysis*

The preliminary analysis of the data for the year 1997 is shown in table 2 and the relevant pair-wise correlations between various ratio groups for this year are shown in table 4. Likewise, preliminary data analysis and the pair-wise correlation matrix for 2003 data are presented, respectively, in tables 3 and 5. These tables show that most correlations are highly significant, suggesting the presence of high collinearity that might produce unstable results. Therefore, PCA, which is an alternative method of statistical analysis, is employed to obtain principal components that can be used for predicting firm growth using an empirical model.

2.3. *Empirical analysis*

2.3.1. Principal component analysis for financial factors. PCA converts the original set of correlated variables into a set of uncorrelated variables. Pearson (1901) originally proposed the notion of principal components that was eventually developed by Hotelling (1933). This technique explains the variance–covariance structure of a large set of collinear variables through a relatively few uncorrelated linear combinations (Dunteman 1989, Jolliffe 2002), known as principal components.

Principal components are linear combinations of the realization of q random variables, $\varphi_1, \varphi_2, \ldots, \varphi_q$, over a sample of n individuals. If we denote the q components by $\phi_1, \phi_2, \ldots, \phi_q$, we see that they form a new representation of the initial n points, achieved by replacing the original (canonical) coordinate system with a new q-dimensional coordinate system whose new coordinate axes is denoted by the basis $\beta_1, \beta_2, \ldots, \beta_q$. These axes represent the direction, variability, as well as the description of the variance structure, enabling components to exclusively depend on the covariance matrix (Σ), or the correlation matrix (P) of the original variables. Thus, assuming that $\Psi = [\varphi_1, \varphi_2, \ldots, \varphi_k]$ is a random matrix with its covariance matrix (Σ) encompassing eigenvalues of the form $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_q \geq 0$,

Table 2. Summary statistics: 1997 sample.

Variable	Mean	Standard deviation	Kurtosis	Skewness	Minimum	Maximum	Count
SGR	72,794.26	701,582.11	92.99	9.64	−29.22	6,765,856	93
TA	29,773,040.73	34,082,547.97	6.61	2.39	1,008,571	185,047,931	93
NS	16,443,826.91	17,863,539.93	8.81	2.70	550,375	94,827,288	93
ROA	9.19	8.42	5.19	1.88	−5.49	44.03	93
ROE	13.61	14.77	4.37	1.44	−24.3	76.97	93
GM	17.09	8.81	1.24	0.89	−3.03	48.01	93
OI	7.05	7.733	1.71	0.83	−11.96	35.26	93
CR	184.29	103.48	4.61	2.04	53.57	558.50	93
ATR	121.07	95.81	5.91	2.27	13.6	510.9	93
DER	85.98	48.16	6.99	2.16	18.64	320.3	93
TIE	42.93	284.80	90.63	9.47	−4.78	2743	93
DAR	63.65	31.94	−0.08	0.48	12.45	163.57	93
DI	105.07	200.92	33.26	5.65	14.29	1365.07	93
DAP	56.53	139.40	86.98	9.18	4.2	1365.07	93
WC	3,628,172.49	4,792,575.93	3.19	1.28	−10,970,111	21,002,031	93
WCA	0.29	0.42	11.26	2.58	−0.67	2.34	93
LTL	0.28	0.14	1.04	0.89	0.06	0.75	93
SLNW	1.42	1.05	1.45	1.36	0.28	4.85	93

Definitions of the variables given in table 1. The sales growth rate (SGR) mean and standard deviation are affected by two extreme outliers (one company's sales growth rate was over 6.7 million percent and the other had a sales growth rate of over 2500%). Elimination of these outliers only strengthens our conclusions.

Table 3. Summary statistics: 2003 sample.

Variable	Mean	Standard deviation	Kurtosis	Skewness	Minimum	Maximum	Count
SGR	61.82	309.27	70.89	8.10	−69.93	2941.92	114
TA	51,117,105.24	65,109,864.07	7.54	2.58	1,453,684	370,015,511	114
NS	31,903,897.33	38,779,763.50	8.71	2.58	507,084	245,008,930	114
ROA	2.05	11.63	8.08	−1.04	−57.75	50.64	114
ROE	−4.36	34.10	6.93	−2.16	−176.4	66.1	114
GM	10.69	12.01	3.56	−1.07	−31.16	49.71	114
OI	−0.09	17.14	10.92	−2.74	−86.39	43.11	114
CR	164.09	98.92	2.28	1.27	14.31	513.28	114
ATR	117.59	86.52	2.18	1.34	407.47	13,405.76	114
DER	179.70	369.59	46.24	6.29	13.91	3255	114
TIE	126.74	762.80	64.49	7.84	−17.78	6958	114
DAR	65.27	39.16	6.51	2.03	7.27	242.80	114
DI	131.82	527.24	51.24	7.16	0.86	4003.47	114
DAP	48.44	29.33	2.51	1.29	6.42	159.3	114
WC	5,515,512.79	11,927,336.73	9.74	2.49	−14,559,165	65,708,919	114
WCA	0.01	0.64	16.86	−3.59	−3.58	0.97	114
LTL	0.30	0.18	0.27	0.87	0.07	0.85	114
SLNW	2.05	2.43	10.49	3.11	0.25	13.88	114

For list of variables, see table 1.

and $\Phi = [\phi_1, \phi_2, \ldots, \phi_q]$, the linear combinations are shown by the following equations:

$$\begin{aligned} \varphi_1 &= \beta_1'\Phi = \beta_{11}\phi_1 + \beta_{12}\phi_2 + \cdots + \beta_{1q}\phi_q, \\ \varphi_2 &= \beta_2'\Phi = \beta_{21}\phi_1 + \beta_{22}\phi_2 + \cdots + \beta_{2q}\phi_q, \\ &\vdots \\ \varphi_q &= \beta_q'\Phi = \beta_{q1}\phi_1 + \beta_{q2}\phi_2 + \cdots + \beta_{qq}\phi_q. \end{aligned} \quad (1)$$

The variance covariance matrix Φ in the above equation is shown respectively in the following equations:

$$\mathrm{Var}(\varphi_i) = \beta_i'\Sigma\beta_i, \quad \text{for } i = 1, 2, \ldots, q, \quad (2)$$

$$\mathrm{Cov}(\varphi_i, \varphi_j) = \beta_i'\Sigma\beta_j, \quad \text{for } i, j = 1, 2, \ldots, q, \quad (3)$$

where these components show that $\mathrm{Var}(\varphi_i) = \beta_i'\Sigma\beta_i$ can increase substantially by multiplying the new coordinate axes by a large constant. Therefore, to eliminate this indeterminacy, it would be appropriate to restrict the coefficient vector to unit length, which would allow defining various principal components adequately. Consequently, the first principal component is the linear combination $\beta_1'\Phi$ that maximizes $\mathrm{Var}(\beta_1'\Phi)$ subject to$\beta_1'\beta_1 = \|\beta_1\|^2 = 1$ and $\mathrm{Cov}(\beta_1'\Phi, \beta_2'\Phi) = 0$. Similarly, the ith component is a linear combination $\beta_1'\Phi$ that maximizes $\mathrm{Var}(\beta_i'\Phi)$ subject to the condition $\beta_i'\beta_i = 1$ and $\mathrm{Cov}(\beta_i'\Phi, \beta_j'\Phi) = 0$ for$i > j \geq 1$.

After obtaining the q principal components where $q < n$, the components that represent most of the sample variation are selected and held for additional analysis

Table 4. Correlation matrix for financial ratios: 1997 sample.

	SGR	TA	NS	ROA	ROE	GM	OI	CR	ATR	DER	TIE	DAR	DI	DAP	WC	WCA	LTL	SLNW
SGR	1.0000																	
TA	−0.0834	1.0000																
NS	−0.0568	**0.8544*****	1.0000															
ROA	**0.2866*****	−0.0269	**0.0941*****	1.0000														
ROE	**0.3409*****	−0.0255	0.1085	**0.9687*****	1.0000													
GM	−0.0283	**0.2291****	0.0898	**0.4665*****	**0.4387*****	1.0000												
OI	0.0351	**0.2346****	**0.1755***	**0.5489***	**0.5160***	**0.7752***	1.0000											
CR	0.0612	−0.0629	−0.0469	**0.4317*****	**0.3250*****	**0.3312*****	**0.3648*****	1.0000										
ATR	0.0632	−0.0621	−0.0304	**0.4947*****	**0.3871*****	**0.3271*****	**0.3507*****	**0.9328*****	1.0000									
DER	−0.0465	−0.0183	0.0426	**−0.3731*****	**−0.3283*****	**−0.3612*****	**−0.2322****	**−0.4312*****	**−0.3806*****	1.0000								
TIE	−0.0039	−0.0272	0.0358	**0.4920*****	**0.3429*****	**0.1864***	**0.3027*****	**0.4103*****	**0.4221*****	−0.1706	1.0000							
DAR	−0.1261	**−0.2992*****	**−0.3170*****	−0.1033	−0.0901	−0.0190	−0.1179	0.0447	0.0948	0.1377	−0.1713	1.0000						
DI	−0.0347	−0.0029	−0.1264	−0.1676	−0.1490	0.1087	0.1211	0.0407	**−0.2134****	0.0212	−0.0351	0.0449	1.0000					
DAP	−0.0032	−0.0169	−0.0831	−0.0164	0.0052	0.1196	0.1047	0.0579	−0.0931	−0.0394	0.0008	−0.0220	**0.6362*****	1.0000				
WC	−0.0508	**0.4961*****	**0.4585*****	**0.2782*****	**0.2212****	**0.3473*****	**0.3685*****	**0.5533*****	**0.4809*****	**−0.2448****	**0.2967*****	**−0.1180**	**0.2667****	**0.1963***	1.0000			
WCA	−0.0231	−0.0110	−0.1436	0.1466	0.1129	**0.3640*****	**0.3049*****	**0.5789*****	**0.3747*****	**−0.2740*****	0.1345	0.1366	**0.7434*****	**0.5110*****	**0.5749*****	1.0000		
LTL	0.0539	**−0.3817*****	**−0.1982***	−0.1092	−0.0490	**−0.3455*****	**−0.2527****	**−0.4587*****	**−0.4102*****	**0.6619*****	−0.0882	**0.1902***	0.0985	0.0706	**−0.3694*****	**−0.2099****	1.0000	
SLNW	**0.2736*****	**−0.3217*****	0.0134	0.0766	0.1305	**−0.3961*****	**−0.2158****	−0.1464	−0.0871	**0.5296*****	−0.0177	−0.0783	**−0.2267****	−0.0904	**−0.1910***	**−0.2859*****	**0.6807*****	1.0000

For list of variables, see table 1. Significant at the *10%, **5% and ***1% level.

Table 5. Correlation matrix of financial ratios: 2003 sample.

	SGR	TA	NS	ROA	ROE	GM	OI	CR	ATR	DER	TIE	DAR	DI	DAP	WC	WCA	LTL	SLNW
SGR	1.0000																	
TA	0.0778	1.0000																
NS	**0.3304***	**0.7120***	1.0000															
ROA	**0.1606***	**0.1600***	**0.3579***	1.0000														
ROE	0.1577*	**0.1941***	**0.3545***	0.9037	1.0000													
GM	0.0196	**0.2324***	**0.2004***	**0.6057***	**0.5990***	1.0000												
OI	0.0785	**0.1852***	**0.2677***	**0.6844***	**0.6053***	**0.8500***	1.0000											
CR	−0.0266	0.1499	0.1414	**0.3161***	**0.2540***	**0.3437***	**0.3223***	1.0000										
ATR	0.0095	0.1458	0.1826*	0.3322***	0.2608***	0.3310***	0.3307***	0.9433***	1.0000									
DER	−0.0501	−0.1488	**−0.1692***	**−0.4544***	**−0.7348***	**−0.2847***	**−0.1646***	**−0.1913***	**−0.1710***	1.0000								
TIE	−0.0112	−0.0278	0.1086	**0.1679***	**0.1561***	0.0174	0.0414	0.0527	0.0758	−0.0470	1.0000							
DAR	−0.0203	**−0.1874***	**−0.1921***	**−0.1725***	−0.1276	**−0.1975***	**−0.2467***	0.0096	0.0439	−0.0609	−0.0273	1.0000						
DI	−0.0674	−0.0566	−0.1321	**−0.2048***	**−0.2417***	**−0.3591***	**−0.3681***	−0.1024	**−0.2105***	0.0685	−0.0339	**0.3047***	1.0000					
DAP	0.0507	**−0.2117***	−0.0759	−0.0869	**−0.1784***	**−0.2900***	**−0.3606***	**−0.1843***	−0.1391	**0.2497***	0.0984	**0.3867***	**0.4617***	1.0000				
WC	**0.1779***	**0.7340***	**0.6851***	**0.3853***	**0.3928***	**0.3838***	**0.3641***	**0.5049***	**0.5237***	**−0.2968***	0.0930	−0.0843	−0.0725	−0.1478	1.0000			
WCA	0.0636	**0.2120***	**0.2144***	**0.3966***	**0.4825***	**0.5266***	**0.4753***	**0.5183***	**0.4570***	**−0.3604***	0.0583	−0.0226	0.0360	−0.0700	**0.4424***	1.0000		
LTL	0.1264	**−0.3906***	−0.1493	**−0.3096***	**−0.4093***	**−0.5614***	**−0.4442***	**−0.5364***	**−0.4685***	**0.3996***	0.0851	0.1444	**0.3006***	**0.4611***	**−0.3740***	**−0.5787***	1.0000	
SLNW	0.0783	**−0.2520***	−0.0432	**−0.2003***	**−0.4403***	−0.1102	0.0702	0.1070	**0.1597***	**0.7386***	0.0411	**−0.2306***	−0.1267	0.0706	**−0.2030***	−0.1077	**0.3364***	1.0000

For list of variables see table 1. Significant at the *10%, **5% and ***1% level.

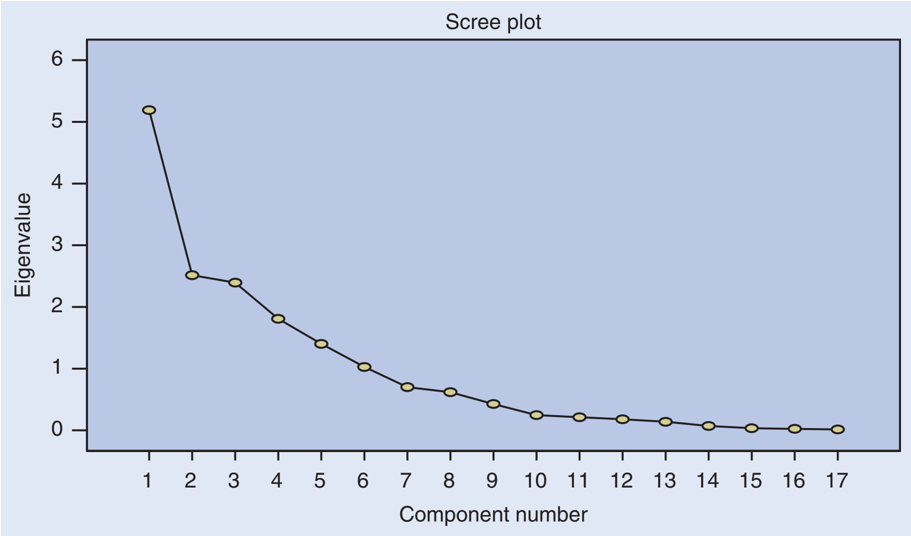

Figure 1. 1997 sample.

using data-retention techniques such as the *scree test*, cross-validation, or Velicer's Partial Correlation Procedure (Jackson 1991). Rousson and Gasser (2004) noted that PCA is not easily interpretable and found that there is a trade-off between optimality and understandability (Vines 2000). Optimality refers to the perseverance of the original variance in the resulting components and even if the new components do not show meaningful patterns, they are rotated using rotational algorithms such as varimax (Kaiser 1958) or quartimax (Harman 1967) so that variances concentrate with sizeable components to provide both optimality as well as interpretability of the results (Saunders 1974). Chiattello (1974) pointed out that rotated components delineate distinct bunches of relationships in the data. Jolliffe and Uddin (2000) reported another alternative that uses simplified component analysis which maintains a balance between these two extremes, but all of the methods assume that the original un-rotated components are difficult to interpret.

Kaiser (1960) recommended retaining only those components whose *eigenvalues* are greater than 1. Jolliffe (1972) recommended selecting components that choose λ to be equal to 0.7 rather than 1. Another component selection method reported by Cattell (1966) is based on the *scree test*, which is a graphical method wherein all components are retained whose eigenvalues lie in the sharp descent of the graph before the first on the line starts to level off. For the first method, one would retain the components with a cumulative variance that would account for at least 70% of the total variance or more, although Joliffe (1986) recommended retaining the first several components with a cumulative variance of between 80 and 90%. However, in the present study, we retain those components that have eigenvalues greater than 1. This selection criterion delivers a cumulative variance above 75%.

2.3.2. Results on principal component analysis. The use of PCA as a data exploratory device allows the reduction of *p* dimensions of complex interrelationships existing in the original data to a few components of distinct dimensions of general patterns. According to Stevens (1973), these distinct components would help us to interpret them, but one would like to know the number of components to be extracted from the PCA and to be retained for subsequent use in the regression analysis. Therefore, to determine the number of components to be retained in our empirical analysis we started with the *scree test*, plots of which are shown in figures 1 and 2, respectively, for the years 1997 and 2003. This criterion suggests retaining six components for each year; however, to minimize the number of components, we decided to use the Kaiser test and retain only those components that had eigenvalues greater than or equal to 1.00. This allowed us to retain six components for 1997 and five components for 2003.

Table 6 shows the eigenvalues, percentages of total variance, cumulative eigenvalues, and cumulative percent of the total variance for all the principal components for the years 1997 and 2003. The table shows that 84% of the variance of the initial variables is accounted for by six components for the year 1997 and 76% of the total variance for the year 2003. This means that over 75% of the variance for each year is accounted for by the first six and five selected ratio groups, respectively, for the years 1997 and 2003. This implies that the remaining variances are either error variance or noise, and not sufficiently dominant to indicate the dimensionality of the common component space. Moreover, results for component 6 for the year 1997 are not tabulated for the reasons given in the subsequent sub-sections as well the later paragraphs of the current sub-section. It should be noted that the information contained in table 6 is based on un-rotated components and thus is not directly comparable to the information provided in tables 7 and 8, where components are rotated.

The un-rotated PCA results did not show any clear interpretable pattern of grouping the various financial ratios. Therefore, these components were rotated using

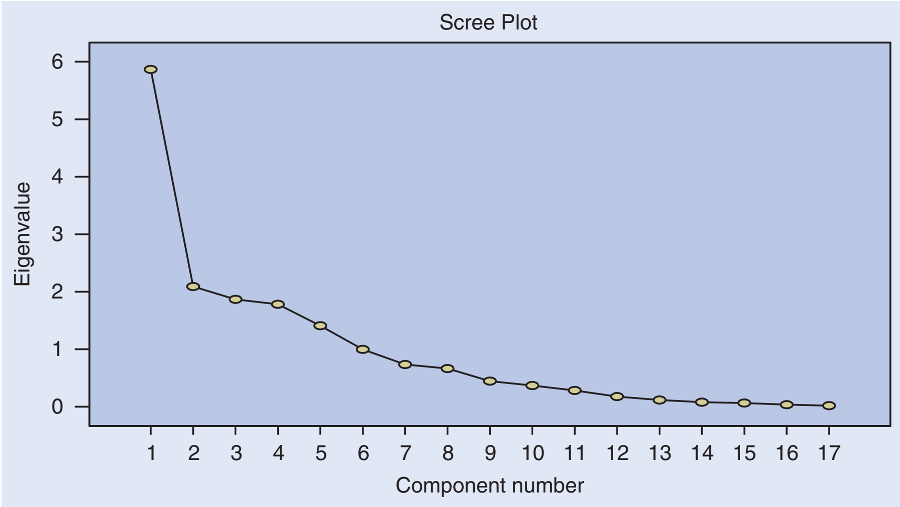

Figure 2. 2003 sample.

Table 6. Unrotated principal components: Eigenvalues and share of variance.

Year	Principal component	Eigenvalue	% of total variance	Cumulative eigenvalue	Cumulative % of variance
1997	Component 1	5.0903	29.9429	5.0903	29.9429
	Component 2	2.5484	14.9909	7.6386	44.9338
	Component 3	2.4221	14.2475	10.0608	59.1813
	Component 4	1.8188	10.6991	11.8797	69.8803
	Component 5	1.3747	8.0862	13.2543	77.9665
	Component 6	1.0396	6.1155	14.2939	84.0820
2003	Component 1	5.8026	34.1331	5.8026	34.1331
	Component 2	2.0882	12.2837	7.8909	46.4168
	Component 3	1.8659	10.9759	9.7568	57.3927
	Component 4	1.7647	10.3803	11.5214	67.7730
	Component 5	1.4453	8.5016	12.9667	76.2745

None of the components have been named since they have not yet been rotated and are thus difficult to interpret.

Table 7. Rotated structure coefficients: 1997 sample.

Variable	ROI	STLIQ	LTSOL	CAPTO	FRMS
TA	0.0385	−0.0876	−0.2128	0.0019	**−0.9403**
NS	0.0770	−0.0021	0.0568	0.1333	**−0.9168**
ROA	**0.8525**	0.3512	0.0385	0.1049	0.0625
ROE	**0.8638**	0.2334	0.0843	0.0937	0.0573
GM	**0.7305**	0.0494	−0.4120	−0.1767	−0.1806
OI	**0.7881**	0.1149	−0.1967	−0.1715	−0.2393
CR	0.1961	**0.9091**	−0.2278	−0.1129	0.0316
ATR	0.2532	**0.8991**	−0.1779	0.1264	0.0304
DER	−0.2443	−0.2682	**0.7586**	0.0029	−0.1589
TIE	0.3229	0.5015	0.0659	−0.0149	0.0358
DAR	−0.0422	0.0887	0.0892	0.0013	0.2653
DI	−0.0611	−0.0692	−0.0284	**−0.9399**	−0.0007
DAP	0.0504	−0.0548	0.0235	**−0.8257**	0.0517
WC	0.1468	0.5900	−0.1206	−0.3462	−0.5883
WCA	0.1116	0.4585	−0.1972	**−0.7874**	0.0033
LTL	−0.019	−0.3074	**0.8494**	−0.0892	0.2405
SLNW	−0.0103	0.0549	**0.8773**	0.1835	0.1096
Explanatory variance	2.9440	2.8308	2.4826	2.4739	2.3391
Proportion of total	0.1732	0.1665	0.1460	0.1455	0.1376

ROI, returns on investments; STLIQ, short-term liquidity; LTSOL, long-term solvency; CAPTO, capital turnover or capital intensiveness; FRMS, a group of financial ratios pertaining to 'firm size. The numbers in this table show the degree of the relationship between a component and a variable. For example, 0.2334 in row 4 column 2 indicates the loadings of variable ROE into component STLIQ. Likewise, the other numbers in the table show similar pair−wise relationships between the other components and variables. Larger positive values indicate that a greater weight is placed on the component by the variable. Negative values indicate an inverse relationship. Thus, while return on assets (ROA) and return on equity (ROE) have a very close relationship with our principal component, return on investment (ROI), as would be expected, and long−term liabilities as a percentage of assets (LTL) show an inverse relationship. Again, this is expected. We use all the variables for each of the five selected components and replace the 17 variables with the five components as a means of data reduction.

Table 8. Rotated structure coefficients: 2003 sample.

Variable	STLIQ	ROI	LTSOL	CAPTO	FRMS
TA	0.0999	−0.0728	−0.1731	0.1611	**0.9056**
NS	0.0246	0.2670	0.0139	0.0562	**0.8765**
ROA	0.2100	**0.8344**	−0.2618	0.1523	0.1393
ROE	0.1746	**0.7486**	−0.5424	0.1758	0.1404
GM	0.3785	0.5658	−0.1910	0.4678	0.0662
OI	0.3292	0.6525	−0.0224	0.5057	0.0797
CR	**0.9415**	0.0783	0.0382	0.0366	0.1014
ATR	**0.9139**	0.1222	0.0997	0.0330	0.1198
DER	−0.1462	−0.2589	**0.8543**	−0.0333	−0.0738
TIE	−0.0449	0.4713	0.1786	−0.2285	0.0797
DAR	0.1642	−0.1465	−0.2302	−0.6408	−0.2195
DI	−0.0615	−0.1603	−0.0778	**−0.7224**	0.0188
DAP	−0.1338	0.1508	0.2060	**−0.8009**	−0.0689
WC	0.4489	0.1859	−0.1390	0.0153	**0.7841**
WCA	0.6359	0.3363	−0.2446	0.0117	0.1302
LTL	−0.6003	0.0010	0.4427	−0.4537	−0.1448
SLNW	0.1057	0.0135	**0.9316**	0.1157	−0.1259
Explanatory variance	3.1081	2.6071	2.4477	2.4072	2.3966
Proportion of total	0.1828	0.1534	0.1439	0.1416	0.1409

See notes for table 7.

the varimax rotation algorithm of Kaiser (1958), which enabled us to obtain meaningful patterns. Consequently, the results encompassing loading matrices are based on the varimax rotation algorithm, with data on 17 financial ratios for Taiwanese firms presented, respectively, in tables 7 and 8 for the years 1997 and 2003.

The factor loadings present the most important information on which the interpretation of factors is based. The similarity of each variable to each of the empirically derived factors is indicated by the variable's factor loading. The factor loading can simply be interpreted as the correlation between the original variable and the particular factor. The factor loading in each factor represents an ordinal measure of the degree to which the variable is involved in that specific component.† Most of the factor interpretations depend on the sign and ordering of the factor weights.

Tables 7 and 8 report the scores for the five rotated components that were later used in the regression analysis. The data in the columns of these tables show the coefficients of each of the components that are linear combinations of all financial variables used in the analysis. When these coefficients are close to zero, they show less association between the specific variable and the component and, likewise, when the coefficient takes on a positive value close to 1.00, the stronger the association between the specific variable and the variable pattern reflected by the component. Alternatively, similar to correlation coefficients, a negative sign on a coefficient in the factor loading matrices would indicate an inverse relationship. We are not interested in determining the relative significance of these relationships because, in doing so, the extracted components would become less useful. Nevertheless, the PCA is itself a data-reduction technique that is used to reduce the 17 variables into five or six components and to alter these dimensions by dropping portions of them would inhibit the validity of the data reduction.

In general, the first vector delineates the most important variable relationship in the data, the second vector the next most important relationship, and so on. Ideally, the financial groups should possess high internal homogeneity within groups and high external heterogeneity between groups; each financial ratio should have a large coefficient value at most on one component.

The component matrix for the year 1997 (table 7) shows that the first principal component represents the largest coefficient value for the variables returns on assets (ROA), returns on equity (ROE), gross margin (GM), and operation income change (OI), and the other variables with a small coefficient value show a low degree of involvement in that particular component pattern. In more specific terms, the first two variables (ROA and ROE) can be labeled as returns on investment (ROI) and the latter two (GM and OI) as profit margin ratios. In more general terms, the first component could be interpreted as the financial dimension of ROI. The second component can be classified in the category of short-term liquidity (STLIQ) since it has a large coefficient value on the variable current ratio (CR) and the variable acid test ratio (ATR). Likewise, the third, fourth and fifth components can be described as the categories 'long-term solvency' (LTSOL), 'capital turnover' (or 'capital intensiveness') (CAPTO), and 'size of corporation' (FRMS).

By examining the factor loadings and the clustering of the financial ratios, the 2003 financial data can be grouped into five identical financial ratio categories with different sequential orders, namely 'short-term liquidity' (STLIQ), 'return on investment' (ROI), 'long-term solvency'

†The total combinations of variables are referred to as vectors, components, factors or eigenvectors.

(LTSOL), 'capital turnover' (CAPTO) and 'size of corporation' (FRMS). We observe that there is a cross-sectional stability of the ratio groups over time in terms of the consistency of the factor loadings and the percent of total variance accounted for by all variable patterns that are linked to specific variable interrelations.

From the results shown in table 7 it transpires that the first component, i.e. ROI, is the most dominant common factor in 1997 and had a variance† of 2.9, accounting for 17% of the total variance, with a high loading of 0.852460 on variable ROA and 0.863803 on variable ROE. However, the results shown in table 8 for the year 2003 reveal that, although ROI became the second most dominant factor, it still had a similar decomposition with a variance of 2.6, 15% of the total variance, a high loading of 0.83448 on variable ROA and 0.748566 on variable ROE. In 1997, the short-term ratio group (STLIQ) was the second most dominant factor and had a variance of 2.8 and accounted for 16% of the total variance, with the highest loading of 0.90912 on variable CR and the second highest loading of 0.899056 on variable ATR. However, in 2003, due to a slight change in decomposition, STLIQ became the most dominant factor, had a variance of 2.8, and 16% of the total variance, with the highest loading of 0.941469 on variable CR and the second highest loading of 0.913962 on variable ATR. Thus, the first two patterns, ROI and STLIQ, are consistent and stable, indicating fairly widespread involvement among cross-sectional corporations in Taiwan between 1997 and 2003.

The remaining three categories of components, namely LTSOL, CAPTO, and FRMS, for both years, i.e. 1997 and 2003, show a similar composition in terms of the size of the factor loadings, the presence of specific financial variables in the same category, and the percent of total variance accounted for by all variable patterns linked to the specific variable interrelation.

In 1997, the LTSOL group was the third most dominant factor, with a variance of 2.4826, which accounted for 14.62% of the total variance, with the highest loading of 0.8773 on variable SLNW, the second highest loading of 0.8494 on variable LTL, and, finally, a loading of 0.7586 on variable DER. Likewise, for the year 2003, LTSOL maintained its position as the third most dominant component, with a variance of 2.447, which accounted for 14.39% of the total variance, with loadings of 0.9316, 0.8543, and 0.4427, respectively, on variables SLNW, DER and LTL.

CAPTO is the fourth most dominant component for the year 1997, and also maintained its position later in year 2003. This component had a variance of 2.4739 in 1997, which was 14.55% of the total variance with factor loadings of −0.9399, −0.8527, and −0.7874, respectively, for variables DI, DAP, and WCA. Such negative factor loadings can be interpreted as this being an *inverted* variable. In other words, the larger the value, the *less* the capital intensiveness. For the year 2003, CAPTO had a variance of 2.4072, which is 14.16% of the total variance, and had factor loadings of −0.7224, −0.8009, and 0.0117, respectively, for variables DI, DAP, and WCA. Although the factor loading for WCA is positive, the overall factor runs negative, maintaining its status as an inverted variable. However, when compared with 1997, in the year 2003 the ratio group CAPTO was identified by the same variables DI with a loading of −0.7224 and variable DAP with a loading of −0.8009, but also had a different variable, WCA, with a low factor loading of 0.0117, indicating a shift between the involvement of variable WCA (working capital/sales) and the derived factor/component (capital intensiveness).

Firm size (FRMS), the fifth component in year 1997, had a variance of 2.3391, accounting for 13.76% of the total variance with factor loadings of −0.9403 and −0.9146, respectively, for variables TA and NS. The ratio group exhibited stability over time and encompassed a variance of 2.3966, which is 14.09% of the total variance with factor loadings of 0.9056 and 0.8765 on variables TA and NS, respectively. This implies that, in 1997, firm size is *inverted* so that the larger the variable, the smaller the firm. However, in 2003, this is reversed, so we need to interpret the factor according to the more traditional definition.

There were a few ratios, such as the variable DAR (turnover rate of accounts receivable), that did not group into any specific dimension of principal components. The reasons for these financial variables not to be grouped into any single category were due to either that the activities of a firm indicated by these ratios could be adequately measured by other financial ratios that have already been grouped into specific categories, or that each of these ratios measured a unique characteristic of a firm's activities and thus lacked empirical similarity to other financial ratios.

In 1997, factor 6 (not reported for brevity) was identified alone by variable DAR, with only one high loading of 0.85016 and extremely low loadings on the others, which explained 6% of the total variance. This financial pattern could possibly be labeled 'Receivable intensiveness', describing the unique financial dimensionality of a company. In 2003, there was no sign of the other remaining factors; variable DAR was not dominant in any of the five components and thus lacked empirical similarity to other financial ratios.

An examination of the components for years 1997 and 2003 shows notable stability in terms of the explanation of variance among all components. Although ROI is the most important component and STLIQ is the second most important component in this regard in 1997, this importance is reversed in 2003. However, the other three components, LTSOL, CAPTO, and FRMS, retain their level of importance in both years. Stability can also be witnessed within components, although there are some notable exceptions. In the case of the size of the corporation, the variable corresponding to long-term

†The amount of variance explained by each rotated component is the sum of the squared loadings shown in the column.

liability/total assets has a significant component value in 1997 only. The component for working capital/total assets drops out in 2003, although it is an important component value in 1997. This indicates a downward shift between the involvement of working capital/total assets, and the derived component, i.e. capital intensiveness. This may have been because other financial ratio groups could adequately measure the activities of a firm or these ratios may be measured by a unique characteristic of a firm. However, the results indicate that the variables capital intensiveness and receivables intensiveness are the least-stable financial ratio patterns. Therefore, the ability to generate and maintain a stable amount of cash flow was challenging for Taiwanese firms, especially when the firms of this small newly industrializing economy are aimed at export-led growth.

2.3.3. Empirical model for financial factors. In order to investigate the impact of financial factors on firm growth in Taiwan, our regression model uses growth as the dependent variable and various financial ratios/components, isolated through exploratory principal component analysis (PCA), as independent variables to study which ratio group (if any) would be able to explain firm growth in Taiwan. Although it would be possible to break out individual variables based on PCA, we chose not to do so since we are concerned with the general category factors that might explain growth in Taiwan rather than focusing on specific variables in our dataset.

Table 7 shows the scores for the five components for which we propose an interpretation and that are later used in the regression analysis. The selected components are employed to estimate firm growth using the model in a general form, which is shown by the following equation:

$$G_{it} = \sum_{j=1}^{m_t} (\gamma_{jt} + \beta_{jt} component_{ijt}) + \varepsilon_{it}, \tag{4}$$

where G_{it} represents the growth rate of firm i, for $t \in \{1997, 2003\}$, and $component_{ijt}$ represents the jth component for firm i that is obtained from the exploratory principal component analysis performed separately for the two available years. γ_{jt} and β_{jt} are the model coefficients, in particular γ_{jt} can be interpreted as fixed effects specific to the relationship between the various independent variables ($component_{ijt}$) of the model and the dependent variable, which is firm growth (G_{it}) in Taiwan. m_t is the number of components retained for the two periods. m_{1997} is equal to 6 and m_{2003} is equal to 5, representing the number of retained components in each case. However, the sixth component for 1997 is not reported in the regression results for brevity.

These components were employed as independent variables in separate regressions to estimate Taiwanese firm growth for the respective years. Therefore, the above equation (equation (4)) is modified to take on the form of equation (5), which can be used to employ all the available ratio groups to predict firm growth in Taiwan:

$$\begin{aligned} G_{it} = \gamma_{it} &+ \beta_{1t} ROI_{it} + \beta_{2t} STLIQ_{it} + \beta_{3t} LTSOL_{it} \\ &+ \beta_{4t} CAPTO_{it} + \beta_{5t} FRMS_{it} + \varepsilon_{it}, \end{aligned} \tag{5}$$

where G_{it} denotes the firm growth rate (where i denotes the ith firm in a particular year and t denotes one of the two years being studied, i.e. 1997 and 2003), and the independent variables are the rotated components, of which the first independent variable *ROI* is a synthetic measure of returns on investment, the second independent variable, short-term liquidity (STLIQ), then comes long-term solvency (LTSOL), capital turnover (CAPTO), and finally the fifth independent variable in the above regression equation is represented by financial ratios pertaining to firm size (FRMS). For the dependent variable, the sales growth rate (SGR) as given in the database is the year-on-year growth in overall sales as expressed in New Taiwanese dollars.

3. Empirical results

3.1. Results from financial factors for firm growth

The regression results with reference to Taiwanese firms for the year 1997 show that ROI is positively and significantly related to firm growth (G_{it}) with a coefficient of 0.2033 at the 5% level of significance. Once two outliers are eliminated (see table 10), ROI increases to 0.5817 and becomes significant at all levels of significance. However, for the year 2003, the coefficient of ROI is −0.0611, which is insignificant and negatively related to firm growth. This shows that firm growth in Taiwan is not a random process and is financed by increasing firm profitability.

Regression results for the year 1997 (table 9) reveal that the ratio group ROI is positively and significantly related to firm growth ($G_{i,1997}$). In addition, although not significant, the ratio groups STLIQ and FRMS are also positively related to firm growth in Taiwan. However, once two outliers are eliminated, FRMS becomes significantly related to firm growth. FRMS is an inverted factor in 1997, such that the larger its value, the smaller the firm. Thus, smaller firms grew faster in 1997 than did larger firms. In addition, LTSOL is negatively related to firm growth in 1997 and is significant at the 10% level once the outliers are removed. This indicates that more highly leveraged firms grew faster for the year 1997, exactly what one would expect in a financing bubble. In 2003, LTSOL is insignificant but positively related to firm growth, indicating that high leverage did not assist firms to grow during that period.

Of the five components that are employed as independent variables for the year 2003, STLIQ and FRMS are statistically significant at the conventional level (5%) of significance. The regression results for the year 2003 show that FRMS is an important ratio group after the 1997 financial crises. This is evident from the parameter estimate of this variable, which is positively and

Table 9. Regression results for selected factors.

Year	Ratio group	Beta	*t*-Statistic	*p*-Value
1997	ROI	**0.2033**	**1.9945**	**0.0493**
	STLIQ	0.0477	0.4679	0.6410
	LTSOL	−0.1530	−1.5012	0.1369
	CAPTO	−0.0449	−0.4401	0.6609
	FRMS	0.1151	1.1288	0.2622
2003	STLIQ	**0.1829**	**1.9897**	**0.0492**
	ROI	−0.0609	−0.6618	0.5095
	LTSOL	0.0611	0.6647	0.5077
	CAPTO	−0.0358	−0.3892	0.6979
	FRMS	**0.2103**	**2.2865**	**0.0242**

For 1997, $R^2=0.08228$. For 2003, $R^2=0.08644$. ROI, return on investment; STLIQ, short-term liquidity; LTSOL, long-term solvency; CAPTO, capital turnover or capital intensiveness; FRMS, a group of financial ratios pertaining to 'firm size'. CAPTO is an inverted variable in both years. Thus, higher levels of CAPTO indicate lower levels of capital turnover. FRMS is an inverted variable in 1997 and a non-inverted variable in 2003. Thus, higher levels of FRMS indicate *smaller* firms in 1997 but *larger* firms in 2003. In this table, column 1 shows the sample periods, column 2 the selected ratio group for each sample period, column 3 the beta for each ratio group from each of the samples and column 4 presents *p*-values for parameter estimates in the same row in the preceding column. The numbers of observations employed in the analysis are 93 for year 1997 and 114 for year 2003. There are two outliers that dramatically weaken the explanatory power in 1997. When these outliers are eliminated (as shown in table 10), the results are strengthened for ROI with no alternation in the significance of the other variables. Elimination of these companies does not materially affect the PCA results.

Table 10. Regression results for selected factors (outliers eliminated).

Year	Ratio group	Beta	*t*-Statistic	*p*-Value
1997	ROI	**0.581682**	**16.8132**	**0.0000**
	STLIQ	0.009676	0.1135	0.9099
	LTSOL	*−0.165547*	*−1.9394*	*5.0558*
	CAPTO	0.09873	1.1577	0.2503
	FRMS	*3.167994*	*4.9692*	*5.0522*

$R^2=0.38919$. See also notes for table 9. There are two outliers in the sales growth rate (one of which has a sales growth rate of 2500% and the other that has a sales growth rate of over 6,000,000%) that dramatically weaken the explanatory power in 1997. When these outliers are eliminated, the results are strengthened. In this case, nearly 39% of the variance can be explained by the model. Outlier elimination also flips the sign on capital turnover. Firm size is positively related to firm growth and this relationship is significant at the 10% level for firm growth, while long-term solvency is negatively related to firm growth and this relationship is significant at the 10% level.

statistically significantly related to firm growth at the conventional level of significance. Since FRMS is a non-inverted factor in 2003, these results suggest that large firms were more successful in financing their growth. These results are in line with Boubaker *et al*. (2008), who demonstrated that firm size and value creation are positively related. This raises the important question of whether or not large firms grow faster than small firms, since small firms grew faster in 1997 but large firms grew faster in 2003. The key to understanding this development lies in LTSOL. Small firms grew faster but ended up doing so with borrowed money. When they overextended, they collapsed. This is in sharp contrast to much of the rest of East Asia where it was the larger firms that borrowed heavily and faced financial crisis. The validity of the paradigm depends on the fact that firm ownership and management are closely interwoven in Taiwan, which means that ownership is not separated from control. Thus, the conservative corporate structure of Taiwanese established firms funds growth through earned profits rather than external financing (Chen *et al*. 2007a). It may also reflect the ability to attract share funding due to greater exposure through firm size.

Gibrat's Law states that a firm's growth is a random process, and 'Gibrat's Law of proportionate effect' states that the probability of a given proportionate change in size during a specified period is the same for all firms in a given industry, thus the smallest firms have the same chance of growing as the industry's largest firms. Likewise, the 'learning model' of Jovanovic (1982) predicts that the annual growth rate of a firm depends on the accuracy at which managers are able to predict the prices of products. Therefore, larger firms become more efficient over time and there is less room for further improvement in these firms in terms of profitability and growth, leading to a random process for growth, especially among larger firms (although smaller firms might still follow a non-stochastic process). These results conform to most of the empirical literature that employs data from developed countries such as the United States (see, for example, Hall (1987) and Hart and Oulton (1996)). Even in cases where the process does not devolve entirely into randomness, there is a tendency for larger firms to grow more slowly than smaller firms (Evans 1987). However, our results for Taiwanese firms located in a newly industrializing economy show that firm growth increases with the growth in its size, thus providing a counterexample to Gibrat's two laws. These results are in line with Esteves (2007), who studied Gibrat's Law using data from Brazilian manufacturing and service sector firms. However, their results for a sub-sample from large and well-established firms are in sharp contrast.

The results from the 1997 data suggest that, prior to the Asian Financial Crisis, small firms that borrowed heavily were the main ones to grow. However, there has been a change that now favors larger firms that do not churn inventory. This suggests the presence of a more conservative investment climate in Taiwan, favoring growth in large established firms, which is what one would expect after a financial crisis.

3.2. *Discussions of results*

Hamilton *et al*. (2002) used a sample of high-technology Canadian firms to test Gibrat's Law and concluded that it does not hold for their sample; however, they demonstrated that firm size has a significant impact on firm growth, causing smaller firms to grow faster than larger ones. Likewise, Lotti and Santarelli (2001) noted that different industries have different patterns of growth and that the size of firms can change significantly over time, showing that the growth process appears to be non-random, which is again contrary to Gibrat's Law. Of course, it is entirely possible that growth follows a random process but it appears to be non-random, thus Gibrat's Law may be violated because of survivorship bias (Taleb 2001). However, our results do seem to suggest that Gibrat's Law did not apply to Taiwanese firms during the time period in question.

Demirguc-Kunt and Maksimovic (1998) noted that access to external financing can play a significant role in firm growth; however, this proposition was challenged by Manning (2003). Likewise, our results reveal that internal, rather than external, financing is prevalent in Taiwan. This is possibly due to Taiwan's financial conservativeness and underdeveloped capital and equity markets (Chen *et al.* 2007a). This was one of the reasons why Taiwan was not hit as hard by the 1997 Asian Financial Crises as the other economies in the region. The financial health realized by the Taiwanese economy due to its corporate conservativeness could have helped avert the 2007–9 Western Financial crises or any other financial crises of this nature in the future. Nevertheless, firm growth through retained earnings and skillful use of financial derivatives can presumably mitigate future financial crises, other things being equal.

4. Conclusion

This research explores how data on various financial variables consisting of 17 financial variables/ratios contained in firm financial statements can be used to determine the link (if any) between various financial ratio groups and firm growth in Taiwan. However, the financial variables contained in the financial statements are highly correlated with each other; therefore, to account for multicollinearity, principal component analysis is employed, which allows a large number of correlated independent variables to be transformed into a smaller number of interconnected identical uncorrelated components.

The principal components or components obtained from the principal component analysis are hard to interpret; therefore, they are rotated so that they bunch together into five new components for the years 1997 and 2003 that can be interpreted meaningfully as similar financial ratio groups, e.g. short-term liquidity, returns on investments, long-term liquidity, firm size and capital turnover ratio. The results based on rotated principal component analysis show that the financial ratios were stable between the years 1997 and 2003; therefore, they were able to be employed subsequently in a regression analysis to test for firm growth in Taiwan.

The regression results using the five ratio groups discussed above reveal that, in the year 1997, firm growth was positively related to short-term liquidity and returns on investment, but negatively related to long-term solvency. Firms that had lower levels of capital turnover and firms that were smaller grew faster in 1997. By the year 2003, short-term liquidity was positively and significantly related to firm growth and larger firms that had high levels of capital turnover tended to grow faster. These results suggest that smaller, heavily indebted firms tended to grow faster prior to the 1997 Asian Financial crises, while larger, self-financing firms tended to grow faster in the aftermath.

Generally, our results reveal that a country can significantly improve its existing financial system by reducing the likelihood of financial contagion. Heavily indebted firms are especially vulnerable to changes in financial patterns and a financial crisis, which adequately explains the change in determinants of firm growth since the crisis ended. At the same time, since these firms tended to be smaller in Taiwan than in the rest of East Asia, the failure of such firms had less of an impact than the failure of large industrial conglomerates in terms of economic growth. The post-crisis returns suggest that firms that maintained borrowings at a reasonable level ended up with superior growth outcomes.

Taiwan contrasts with other Asian economies due to its political paranoia and financial conservativeness, which might limit the growth of corporations to some extent; however, it seems to have helped Taiwan to combat the financial storm of 1997. In short, when the relationship between long-term solvency and growth changed from negative to positive, Taiwan was better positioned than its neighbors and thus did not suffer economic catharsis to the same extent.

Acknowledgements

The authors wish to thank the three anonymous referees who significantly improved the quality of this paper. All errors and omissions remain the responsibility of the authors.

References

Akerlof, G.A., The market for lemons: Quality uncertainty and the market mechanism. *Q. J. Econ.*, 1970, **84**, 488–500.

Altman, E., Financial ratios, discriminant analysis, and the prediction of corporate bankruptcy. *J. Finance*, 1968, **23**, 589–609.

Arrow, K., The trade-off between growth and equity. In *Theory for Economic Efficiency: Essays in Honor of Abba P. Lerner*, edited by H.I. Greenfield, A.M. Levenson, W. Hamovitch, and E. Rotwein, pp. 1–11, 1979 (MIT Press: Cambridge, MA).

Audretsch, D. and Elston, J., Does firm size matter? Evidence on the impact of liquidity constraints on firm investment behavior in Germany. *Int. J. Ind. Organiz.*, 2002, **20**, 1–17.

Beaver, W., Financial ratios as predictors of failure. *J. Account. Res.*, 1966, **4**, 71–102.

Bond, S., Elston, J., Mairesse, J. and Mulkay, B., Financial factors and the investment in Belgium, France, Germany, and the United Kingdom: A comparison using company panel data. *Rev. Econ. Statist.*, 2003, **85**, 153–165.

Boubaker, A., Mensi, W. and Nguyen, D.K., More on corporate diversification, firm size and value creation. *Econ. Bull.*, 2008, **7**, 1–7.

Burkowitz, D., Pistor, K. and Richard, J.F., Economic development, legality, and the transplant effect. *Eur. Econ. Rev.*, 2003, **47**, 165–195.

Carpenter, R.E. and Petersen, B.C., Is the growth of small firms constrained by internal finance? *Rev. Econ. Statist.*, 2002, **84**, 298–309.

Carroll, J.B., Biquartimin criterion for rotation to oblique simple structure in factor analysis. *Science*, 1957, **126**, 1114–1115.

Cattell, R.B., The scree test for the number of factors. *Multivar. Behav. Res.*, 1966, **1**, 245–276.

Chan, T.W. and Mozurkewich, M., Interactive comment on *Atmos. Chem. Phys. Discuss.*, 2006, **6**, 10493–10522.

Chen, E.H., Kiani, K.M. and Madjd-Sadjadi, Z., Impact of control type on firm growth in Taiwan. *Int. J. Appl. Econ.*, 2007a, **5**(1), 1–9.

Chen, J., Kan, K.L. and Anderson, H., Size, book/market ratio, and risk factor returns: Evidence from China A-share market. *Managerial Finance*, 2007b, **33**(8), 574–594.

Chen, K. and Shimerda, T.A., An empirical analysis of useful financial ratios. *Financial Mgmt.*, 1981, **Spring**, 51–60.

Chiattello, M.L., On the use of principal component analysis to interpret cross sectional differences among commercial banks: A comment. *J. Financial Quant. Anal.*, 1974, December.

Chittenden, F., Hall, G. and Hutchinson, P., Small firm growth, access to capital markets and financial structure: Review of issues and an empirical observation. *Small Business Econ.*, 1996, **8**(1), 59–67.

Cybinski, P., Description, explanation, prediction – the evolution of bankruptcy studies. *Managerial Finance*, 2001, **27**(4), 29–44.

Demirguc-Kunt, A. and Maksimovic, V., Law, finance, and firm growth. *J. Finance*, 1998, **53**, 2107–2137.

Diambeidou, M.B., François, D., Gailly, B., Verleysen, M. and Wertz, V., Empirical taxonomy of start-up firms' growth trajectories. In *The Dynamics Between Entrepreneurship, Environment and Education*, edited by A. Fayolle and P. Kyrö, pp. 193–219, 2008 (Edward Elgar: Cheltenham).

Dunteman, G., *Principal Components Analysis*, 1989 (Sage: Newbury Park, CA).

Durnev, A. and Kim, E.H., To steal or not to steal: Firm attributes, legal environment, and valuation. *J. Finance*, 2005, **60**, 1461–1493.

Esteves, L.A., A note on Gibrat's law, Gibrat's legacy and firm growth: Evidence from Brazilian companies. *Econ. Bull.*, 2007, **12**, 1–7.

Evans, D.S., The relationship between firm growth, size, and age: Estimates for 100 manufacturing industries. *J. Ind. Econ.*, 1987, **35**(4), 567–581.

Fagiolo, G. and Luzzi, A., Do liquidity constraints matter in explaining firm size and growth? Some evidence form the Italian manufacturing industry. *Ind. Corp. Change*, 2006, **15**, 1–39.

Glancey, K., Determinants of growth and profitability in small entrepreneurial firms. *Int. J. Entrepreneur. Behav. Res.*, 1998, **4**(1), 18–27.

Greene, W.H., *Econometric Analysis*, 2003 (Prentice-Hall: Upper Saddle River, NJ).

Grice, J.S. and Ingram, R.W., Tests of the generalizability of Altman's bankruptcy prediction model. *J. Business Res.*, 2001, **54**, 53–61.

Gulati, D. and Zantout, Z., Inflation, capital structure, and immunization of the firm's growth potential. *J. Financial Strateg. Decis.*, 1997, **10**(1), 77–90.

Hall, B.H., The relationship between firm size and firm growth in the US manufacturing sector. *J. Ind. Econ.*, 1987, **35**(4), 583–606.

Hamilton, O., Shapiro, D. and Vining, A., The growth patterns of Canadian high-tech firms. *Int. J. Technol. Mgmt*, 2002, **24**, 458–472.

Harman, H.H., *Modern Factor Analysis*, 1967 (University of Chicago Press: Chicago).

Hart, P.E. and Oulton, N., Growth and size of firms. *Econ. J.*, 1996, **106**(438), 1242–1252.

Heshmati, A., On the growth of micro and small firms: Evidence from Sweden. *Small Business Econ.*, 2001, **17**(3), 213–228.

Hotelling, H., Analysis of a complex of statistical variables into principal components. *J. Educ. Psychol.*, 1933, **24**, 417–441.

Hutchinson, J. and Xavier, A., Comparing the impact of credit constraints on the growth of SMEs in a transition economy with an established market economy. *Small Business Econ.*, 2006, **27**(2), 169–179.

Jackson, J.E., *A User's Guide to Principal Components*, 1991 (Wiley: New York).

Johnson, R.A. and Wichern, D.W., *Applied Multivariate Statistical Analysis*, 2002 (Prentice-Hall: Upper Saddle River, NJ).

Jolliffe, I.T., Discarding variables in principal component analysis. *Appl. Statist.*, 1972, **21**, 160–173.

Jolliffe, I.T., *Principal Component Analysis*, 1986 (Springer: New York).

Jolliffe, I.T., *Principal Component Analysis*, 2002 (Springer: New York).

Jolliffe, I.T. and Uddin, M., The simplified component technique: An alternative to rotated principal components. *J. Comput. Graph. Statist.*, 2000, **9**, 689–710.

Jovanovic, B., Selection and the evolution of industry. *Econometrica*, 1982, **50**, 649–670.

Kaiser, H.F., The varimax criterion for analytic rotation in factor analysis. *Psychometrika*, 1958, **23**, 187–200.

Kaiser, H.F., The application of electronic computers to factor analysis. *Educ. Psychol. Measure.*, 1960, **20**, 141–151.

Kiani, K.M. and Bidarkota, P.V., On business cycle asymmetries in G7 countries. *Oxford Bull. Econ. Statist.*, 2004, **66**(3), 333–351.

LeClere, M.J., Bankruptcy studies and *ad hoc* variable selection: A canonical correlation analysis. *Rev. Account. Finance*, 2005, **5**(4), 410–422.

Lotti, F. and Santarelli, E., Is firm growth proportional? An appraisal of firm size distribution. *Econ. Bull.*, 2001, **12**, 1–7.

Manning, M.J., Finance causes growth: Can we be so sure? Contributions to *Macroeconomics*, 3: Article 12 (2003).

Meador, A.L., Church, P.H. and Rayburn, L.G., Development of prediction models for horizontal and vertical mergers. *J. Financial Strateg. Decis.*, 1996, **9**, 11–23.

Miller, M.H. and Modigliani, F., Dividend policy, growth, and the valuation of shares. *J. Business*, 1961, **33**(4), 411–433.

Miller, M.H., Debt and taxes. *J. Finance*, 1977, **32**(2), 261–275.

Modigliani, F. and Miller, M.H., The cost of capital, corporation investment, and the theory of investment. *Am. Econ. Rev.*, 1958, **48**, 261–297.

Modigliani, F. and Miller, M.H., Corporate income taxes and the cost of capital: A correction. *Am. Econ. Rev.*, 1963, **53**(3), 433–443.

Musso, P. and Schiavo, S., The impact of financial constraints on firm survival and growth. *J. Evol. Econ.*, 2008, **18**, 135–149.

Myers, S.C. and Majluf, N.F., Corporate financing and investment decision when firms have information that investors do not have. *J. Financial Econ.*, 1984, **13**, 187–221.

Oliveira, B. and Fortunato, A., Firm growth and liquidity constraints. A dynamic analysis. *Small Business Econ.*, 2006, **27**, 139–156.

Padachi, K., Trends in working capital management and its impact on firms' performance: An analysis of Mauritian small manufacturing firms. *Int. Rev. Business Res. Pap.*, 2006, **2**(2), 45–58.

Pearson, K., On lines and planes of closest fit to system of points in space. *Phil.*, 1901, **May 02**, 559–572.

Pinches, G.E. and Mingo, K.A., A multivariate analysis of industrial bond ratings. *J. Finance*, 1973, **28**, 1–18.

Pomerleano, M., The East Asia crisis and corporate finances: The untold micro story. *Emerg. Mkts Q.*, 1998, **2**, 14–27.

Romero, I., Component selection for principal component analysis-based extraction of arterial fibrillation. *Comput. Cardiol.*, 2006, **33**, 137–114.

Rosson, V. and Gasser, T., Simple component analysis. *J. R. Statist. Soc. Ser. C*, 2004, **53**, 539–555.

Saunders, R.J., Further comment: Cross-sectional differences among commercial banks. *J. Financial Quant. Anal.*, 1974, **9**, 1053–1055.

Schatzberg, J.D. and Weeks, D., Security choice, information effects and firm characteristics: A factor analytic approach. *J. Business Finance Account.*, 2004, **31**(9/10), 1483–1503.

Smyth, D.J., Boyles, W.J. and Peseau, D.E., *Size, Growth, Profits and Executive Compensation in the Large Corporation*, 1975 (Macmillan: London).

Spathis, C., Doumpos, M. and Zopounidis, C., Detecting falsified financial statements: A comparative study using multicriteria analysis and multivariate statistical techniques. *Eur. Account. Rev.*, 2002, **11**, 509–535.

Stevens, D.L., Financial characteristics of merged firms: A multivariate analysis. *J. Financial Quant. Anal.*, 1973, **l8**, 149–158.

Taleb, N.N., *Fooled by Randomness: The Hidden Role of Chance in the Markets and in Life*, 2001 (Texere: New York).

Vines, S.K., Simple principal components. *Appl. Statist.*, 2000, **49**, 441–451.

Wiklund, The sustainability of the entrepreneurial orientation–performance relationship. In *Entrepreneurship and the Growth of Firms*, edited by P. Davidsson, F. Delmar and J. Wiklund, pp. 141–155, 2006 (Edward Elgar: Cheltenham).

Yusdar, H., Rahim, A.A., Musa, M.H. and Hashim, A., Principal component analysis of factors determining phosphate rock dissolution on acid soils. *Indones. J. Agric. Sci.*, 2007, **8**, 10–16.